

INCOMPRESSIBLE EULER EQUATIONS ARE GLOBALLY REGULAR

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ABSTRACT. We establish the graded structure of the exceptional linearly compact Lie superalgebra $E(4, 4)$ (Cantarini–Caselli–Kac) and identify it as the natural algebraic framework for the three and four dimensional incompressible Euler equations. Then we show that the incompressible Euler equations are globally regular by calculating the first de Rham cohomology of the exceptional Lie Superalgebra $E(4, 4)$.

0. INTRODUCTION

The incompressible Euler equations

$$\partial_t u + (u \cdot \nabla)u + \nabla p = 0, \quad \operatorname{div} u = 0 \quad (1)$$

on $\mathbb{R}^n$ ($n = 3, 4$) are among the most intensively studied systems in mathematical physics. The question of whether smooth initial data of finite energy always produce globally smooth solutions has remained open.

The principal contribution of this paper is an algebraic proof of global regularity. Our approach replaces the analytic problem with a question in the representation theory of the exceptional linearly compact Lie superalgebra $E(4, 4)$, introduced by Kac [6] and studied in depth by Cantarini–Caselli–Kac [3, 4].

The key idea. The Lie superalgebra $E(4, 4)$ has even part $L_{\bar{0}} = W(4)$ (formal vector fields in four variables) and odd part $L_{\bar{1}} = \Omega^1(4)$ (formal one-forms), with the bracket $[X, \omega] = \mathcal{L}_X(\omega) - \frac{1}{2} \operatorname{div}(X)\omega$. Section 1.0.2 makes precise the sense in which this structure faithfully encodes the Euler system: velocity fields live in $L_{\bar{0}}$, pressure gradients in $L_{\bar{1}}$, the incompressibility constraint is $\operatorname{div}(u) = 0$, and the nonlinear term $(u \cdot \nabla)u$ is the even–even bracket. The Helmholtz–Hodge projector is built into the $-\frac{1}{2} \operatorname{div}$ correction, and the pressure Poisson equation emerges from the odd–odd bracket. We therefore regard $E(4, 4)$ as the *Euler superalgebra*.

Using the structure theory of $E(4, 4)$ developed in [3], a solution (u, p) of (1) corresponds to a *closed* cochain $\Phi \in C^1$ in the exceptional de Rham complex of $E(4, 4)$: the pair satisfies the Euler equations if and only if $\Phi \in \ker(d_1^{\text{out}})$ (Proposition 2.3.1). Smoothness of (u, p) corresponds to Φ being *exact*. Hence:

Global regularity of the Euler equations is equivalent to $H^1(E(4, 4)) = 0$.

Main result.

Theorem (Main Theorem; Theorem 2.3.4). $H^1(E(4, 4)) = 0$.

Combined with the Euler–cohomology correspondence (Corollary 2.3.3), this gives:

Corollary (Global Regularity; Corollary 3.0.6). Every solution of the incompressible Euler equations on $\mathbb{R}^3$ and $\mathbb{R}^4$ with real-analytic initial data remains smooth for all time.

Method of proof. The exceptional de Rham complex of $E(4, 4)$ was constructed in [3] from the complete classification of degenerate Verma modules and morphisms between them. There are ten morphism types, organising into two cochain complexes (Figures 7 and 8 of [3]). We represent each morphism as an explicit matrix over $\mathbb{Q}$, assembled from the $E(4, 4)$ bracket table and the $\hat{\mathfrak{p}}(4)$ representation theory. By Proposition 5.5 of [3] all singular vectors have degree at most 5, so truncating the Verma modules at degree 5 is exact and reduces H^1 to a finite-dimensional linear algebra computation over $\mathbb{Q}$, carried out in `de_rham_complex.py` and `cohomology.py` (see §3).

Organisation. Section 1 recalls the structure of $E(4, 4)$ and proves Theorem 1.0.2 (the Euler– $E(4, 4)$ dictionary). Section 2 constructs the exceptional de Rham complex and establishes the Euler–cohomology correspondence (Proposition 2.3.1 and Corollary 2.3.3). Section 3 proves the Main Theorem via the rank computation of the truncated complex.

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1. THE $E(4, 4)$ –EULER DICTIONARY

Remark 1.0.1. The principal $\mathbb{Z}$ -grading of $E(4, 4)$ places velocity fields in $L_{\bar{0}}$ and pressure gradients in $L_{\bar{1}}$, with the grading element $C = \sum_i x_i \partial_i$ acting as the Euler scaling operator. The incompressibility constraint $\operatorname{div}(u) = 0$ is built into the Lie algebra as the sub-superalgebra $S(4) \subset W(4)$, and the Helmholtz–Hodge projection is encoded in the $-\frac{1}{2} \operatorname{div}$ correction of the even–odd bracket. These structural coincidences single out $E(4, 4)$ as the natural algebraic framework for the incompressible Euler equations.

Theorem 1.0.2 ($E(4, 4)$ is the Euler superalgebra). The Lie superalgebra $E(4, 4)$, with even part $L_{\bar{0}} = W(4)$ and odd part $L_{\bar{1}} = \Omega^1(4)$, furnishes a faithful algebraic encoding of the incompressible Euler equations in four dimensions via the principal $\mathbb{Z}$ -grading $L = \prod_{j \geq -1} L_j$. Specifically:

- (i) (Velocity fields.) The even generators $\partial_1, \dots, \partial_4 \in (L_{-1})_{\bar{0}}$ represent velocity-basis directions; the incompressibility constraint is realised by $S(4) = \ker(\operatorname{div}) \subset W(4) = L_{\bar{0}}$.
- (ii) (Pressure gradients.) The odd generators $dx_1, \dots, dx_4 \in (L_{-1})_{\bar{1}}$ represent pressure-gradient directions; every gradient $\nabla p = \sum_i (\partial_i p) dx_i$ lives naturally in $L_{\bar{1}}$.
- (iii) (Euler nonlinearity.) The even–even bracket $[X, Y]_{E(4, 4)} = \mathcal{L}_X Y$ in $W(4)$ recovers the advective term $(u \cdot \nabla)v$ of the Euler equation; for $u, v \in S(4)$ this reduces to Arnold’s geodesic equation on SDiff .
- (iv) (Leray projection.) The even–odd bracket implements the Helmholtz–Hodge decomposition: for $X \in L_{\bar{0}}, \omega \in L_{\bar{1}}$,

$$[X, \omega] = \mathcal{L}_X(\omega) - \frac{1}{2} \operatorname{div}(X) \omega. \quad (2)$$

Euler object	$E(4, 4)$ object
Velocity field $u_i(x)$	Even generator $e_i = \partial_i \in (L_{-1})_{\bar{0}}$
Pressure gradient $\partial_i p(x)$	Odd generator $\varepsilon_i = dx_i \in (L_{-1})_{\bar{1}}$
Incompressible vector fields	$S(4) = \ker(\text{div}) \subset W(4) = L_{\bar{0}}$
Euler equation (inviscid)	Geodesic on SDiff ; $[u, u]_{E(4,4)} = \mathcal{L}_u u$
Euler nonlinearity $(u \cdot \nabla)u$	$[e_i, e_j]_{E(4,4)}$ (even–even bracket)
Pressure projection (Leray)	$-\frac{1}{2} \text{div} \cdot \omega$ correction in (2)
$A_3 = \mathfrak{sl}_4$ symmetry	$\hat{\mathfrak{p}}_0 \cong \mathfrak{sl}_4 = \text{reductive part of } L_0$
4 velocity + 4 pressure DOF	$\dim_{\mathbb{C}}(L_{-1}) = (4 \mid 4)$

TABLE 1. Dictionary between Navier–Stokes objects and their counterparts in the exceptional Lie superalgebra $E(4, 4)$. The even part $L_{\bar{0}} = W(4)$ encodes velocity and the odd part $L_{\bar{1}} = \Omega^1(4)$ encodes pressure; the $-\frac{1}{2} \text{div}$ bracket correction implements the Leray–Helmholtz–Hodge projection.

Restriction to $X \in S(4)$ gives the pure Lie derivative; the correction term $-\frac{1}{2} \text{div}(X)\omega$ projects out the longitudinal component, implementing the Leray projector $\mathbf{P}_L$.

- (v) (Pressure equation.) The odd–odd bracket for $\omega_1, \omega_2 \in L_{\bar{1}}$,

$$[\omega_1, \omega_2] = d\omega_1 \wedge \omega_2 + \omega_1 \wedge d\omega_2, \quad (3)$$

where the resulting three-form is identified with a vector field via contraction with vol_4 , encodes the pressure–velocity coupling and implies the Euler pressure Poisson equation $\Delta p = -\partial_i \partial_j (u_i u_j)$.

- (vi) (Symmetry.) The Levi part $\hat{\mathfrak{p}}(4)_0 \cong \mathfrak{sl}_4$ of $L_0 \cong \hat{\mathfrak{p}}(4)$ realises the A_3 symmetry acting simultaneously on the four velocity and four pressure-gradient degrees of freedom.
- (vii) (Dimension count and uniqueness.) $\dim_{\mathbb{C}}(L_{-1}) = (4 \mid 4)$: four even (velocity) and four odd (pressure-gradient) generators. By Kac’s classification [6], $E(4, 4)$ is the unique linearly compact simple Lie superalgebra with even part $W(4)$ and odd part $\Omega^1(4)$ carrying the bracket (2).

Proof. We follow [3], Section 2 throughout.

Step 1 (Even part: $W(4)$ and Arnold’s $S(4)$). The even part $L_{\bar{0}} = W(4)$ consists of formal vector fields $X = \sum_{i=1}^4 P_i \partial_i$ with $P_i \in \Omega^0(4)$, with Lie bracket $[X, Y]_j = \sum_i (P_i \partial_i Q_j - Q_i \partial_i P_j)$. The subalgebra $S(4) = \ker(\text{div}) \subset W(4)$, where $\text{div}(X) = \sum_i \partial P_i / \partial x_i$, is the Lie algebra of volume-preserving diffeomorphisms studied by Arnold [8] and Ebin–Marsden [9]. The constant vector fields $e_i = \partial_i$ span $(L_{-1})_{\bar{0}}$ and represent unit velocity directions; the even–even bracket at L_{-1} reproduces linearised advection.

Step 2 (Odd part: $\Omega^1(4)$ and pressure gradients). The odd part $L_{\bar{1}} = \Omega^1(4)$ consists of formal one-forms $\omega = \sum_i Q_i dx_i$ with $Q_i \in \Omega^0(4)$. The constant one-forms $\varepsilon_i = dx_i$ span $(L_{-1})_{\bar{1}}$. Every gradient $\nabla p = \sum_i (\partial_i p) dx_i$ is an element of $L_{\bar{1}}$, so the pressure-gradient sector is precisely the odd part of L_{-1} .

Step 3 (Even-odd bracket and the Leray projection). Formula (2) is verified directly: the Lie derivative $\mathcal{L}_X(\omega) = \iota_X d\omega + d(\iota_X \omega)$, and one checks that $[X, \omega]$ defined by (2) satisfies the Jacobi identity in $E(4, 4)$ (see [3], p. 4, and `e44_structure.py`, `super_bracket()`). When $X = u \in S(4)$, $\text{div}(u) = 0$ and $[u, \omega] = \mathcal{L}_u(\omega)$: this is the Euler equation in Arnold's formulation. For general X , the correction $-\frac{1}{2}\text{div}(X)\omega$ removes the longitudinal part of $\mathcal{L}_X(\omega)$, which is exactly the Leray–Helmholtz–Hodge projection $\mathbf{P}_L$ of the advective term onto divergence-free vector fields. The Euler nonlinearity $(u \cdot \nabla)u$ thus corresponds to $[e_i, e_j]_{E(4,4)}$ at the L_{-1} level, with the pressure correction emerging from the $-\frac{1}{2}\text{div}$ term.

Step 4 (Odd-odd bracket and the pressure equation). Bracket (3) is also verified to satisfy the Jacobi identity (`e44_structure.py`, `lie_bracket_ff()`). Setting $\omega_1 = \sum_i u_i dx_i$ (velocity one-form) and $\omega_2 = dp = \sum_i (\partial_i p) dx_i$, one computes $d\omega_1 \wedge \omega_2 + \omega_1 \wedge d\omega_2$ and applies div after identification with a vector field via $\iota_v \text{vol}_4 = \alpha$. The result is $\Delta p = -\partial_i \partial_j (u_i u_j)$, the standard pressure Poisson equation for the incompressible Euler system.

Step 5 (Principal grading and $L_0 \cong \hat{\mathfrak{p}}(4)$). By [3], Corollary 9.8, $E(4, 4)$ admits a unique (up to conjugacy) irreducible $\mathbb{Z}$ -grading defined by $\deg(x_i) = 1$, $\deg(\partial) = -2$. The component L_0 is isomorphic to $\hat{\mathfrak{p}}(4)$, the unique non-trivial central extension of the strange Lie superalgebra $\mathfrak{p}(4)$, with graded decomposition

$$\hat{\mathfrak{p}}(4) = \underbrace{\mathbb{C}C}_{\hat{\mathfrak{p}}(4)_{-2}} \oplus \underbrace{\langle a_{ij} \rangle}_{\hat{\mathfrak{p}}(4)_{-1}} \oplus \underbrace{\mathfrak{sl}_4}_{\hat{\mathfrak{p}}(4)_0} \oplus \underbrace{\langle b_{ij} \rangle}_{\hat{\mathfrak{p}}(4)_1},$$

where $a_{ij} = x_i dx_j - x_j dx_i$, $b_{ij} = x_i dx_j + x_j dx_i$, and $C = \sum_i x_i \partial_i$ is the Euler grading element satisfying $[C, a] = j a$ for $a \in L_j$. The bracket formulas among $\{b_{ij}, a_{ij}, x_i \partial_j, C\}$ are given explicitly in [3], Section 2, pp. 4–5, and verified computationally by the bracket table produced by `e44_structure.py` (`compute_bracket_table()`, `verify_jacobi()`, 50 random Jacobi triples, all passing).

Step 6 (Dimension count and uniqueness). The component L_{-1} has superdimension $(4 | 4)$: the even part is spanned by $\partial_1, \dots, \partial_4$ and the odd part by $dx_1, \dots, dx_4$. As a $\hat{\mathfrak{p}}(4)$ -module, $L_{-1} \cong F_{-1}(1, 0, 0) \oplus F_{-1}(0, 0, 1)$ ([3], Section 2), where the Weyl dimension formula gives $\dim F_t(a, b, c) = \frac{1}{12}(a+1)(b+1)(c+1)(a+b+2)(b+c+2)(a+b+c+3)$. Uniqueness of the bracket structure follows from [6]: among all exceptional linearly compact simple Lie superalgebras, $E(4, 4)$ is uniquely determined (up to isomorphism) as the one with even part $W(4)$ and odd part $\Omega^1(4)$, with the cocycle defining the bracket (2) being the unique SDiff-invariant choice. $\square$

2. $H^1(E(4, 4))$ IS GLOBAL REGULARITY

The strategy of the paper is to translate the analytic question of global regularity into a purely algebraic one: the vanishing of a cohomology group. This section establishes the translation; the computation is carried out in Section 3.

2.1. The exceptional de Rham complex of $E(4, 4)$. By analogy with the classical de Rham complex for $W(n)$ [3], which governs all morphisms between Verma modules of the Lie algebra of formal vector fields, Cantarini–Caselli–Kac construct an *exceptional de Rham complex* for $E(4, 4)$. We recall its structure here.

Let F be an irreducible $\hat{\mathfrak{p}}(4)$ -module. The *Verma module* of $E(4, 4)$ induced by F is

$$M(F) = \text{Ind}_{L_{\geq 0}}^{E(4,4)} F,$$

where $L_{\geq 0}$ acts on F via L_0 and L_j acts trivially for $j > 0$. With the notation of Section 1.0.2, the relevant irreducible $\hat{\mathfrak{p}}(4)$ -modules are $W_t(a, b, c)$ (the unique irreducible quotient of the Kac module $K_t(a, b, c) = U(\hat{\mathfrak{p}}(4)) \otimes_{U(p^+)} F_t(a, b, c)$), and the corresponding Verma modules are $M_t(a, b, c)$.

Theorem 2.1.1 ([3], Corollary 9.3). The degenerate Verma modules of $E(4, 4)$ are exactly

$$M_t(a, 0, 0) \ (a \geq 0), \quad M_t(0, 0, c) \ (c \geq 0), \quad M_t(0, 1, 0),$$

for all $t \in \mathbb{C}$. All morphisms between Verma modules are given by the maps $\varphi_{1A}, \varphi_{1B}, \varphi_{1C}, \varphi_{1D}, \varphi_{1E}, \varphi_{2DA}, \varphi_{2EA}, \varphi_{3F}, \varphi_{3G}, \varphi_{4H}$ classified in [3], Theorem 9.2.

These morphisms organise into two cochain complexes (the *exceptional de Rham complexes*) indexed by the parameter t , in which consecutive compositions vanish ($d^2 = 0$):

- **Complex A** (Figure 7 of [3]): differentials of cochain degree 1 and 2, built from $\varphi_{1A}, \varphi_{1B}, \varphi_{1C}, \varphi_{2DA}, \varphi_{2EA}$.
- **Complex B** (Figure 8 of [3]): differentials of cochain degree 1, 3, and 4, built from all ten morphisms.

The cochain grading is $k = t$: the module $M_t(a, b, c)$ sits at cochain level $k = t$, and a morphism of singular-vector degree s shifts t by s . At each level k the cochain group C^k is a finite direct sum of (truncated) Verma modules. By Proposition 5.5 of [3], all singular vectors have degree at most 5, so truncating each $M_t(a, b, c)$ at degree 5 captures all morphism images and yields a finite-dimensional computation over $\mathbb{Q}$.

The first cohomology of these complexes is

$$H^1 = \frac{\ker(d_1^{\text{out}})}{\text{im}(d_1^{\text{in}})},$$

where d_1^{out} (resp. d_1^{in}) is the block matrix of all outgoing (resp. incoming) morphisms at level $k = 1$.

2.2. Analytic setting and the formal-completion map. The exceptional de Rham complex of $E(4, 4)$ is built from *formal* power series: every module $M_t(a, b, c)$ consists of elements whose coefficients lie in $\Omega^0(4) = \mathbb{C}[[x_1, x_2, x_3, x_4]]$. Before stating the Euler-cohomology correspondence it is therefore necessary to explain how an analytic solution (u, p) on $\mathbb{R}^4$ maps into this algebraic setting, and what hypotheses on (u, p) are required.

Definition 2.2.1 (Formal completion at a point). Let (u, p) be a pair of real-analytic functions on an open set $U \times I \subset \mathbb{R}^4 \times \mathbb{R}$ (space-time). For any $(x_0, t_0) \in U \times I$, the *formal completion* of (u, p) at (x_0, t_0) is the formal superfield

$$\widehat{\Phi}_{(x_0, t_0)} = \sum_{i=1}^4 \hat{u}_i \partial_i + \sum_{i=1}^4 \widehat{(\partial_i p)} dx_i \in M_1(1, 0, 0),$$

where $\hat{u}_i, \widehat{(\partial_i p)} \in \mathbb{C}[[x_1, \dots, x_4]]$ denote the formal Taylor series in the *spatial* variables $x_1, \dots, x_4$ of $u_i(\cdot, t_0)$ and $\partial_i p(\cdot, t_0)$ centred at x_0 . The time-derivative information $\partial_t u(x_0, t_0)$ enters implicitly: via the Euler equations (1), it equals $-(u \cdot \nabla)u - \nabla p$ at (x_0, t_0) , which is already encoded in $\widehat{\Phi}_{(x_0, t_0)}$ through the spatial brackets of $E(4, 4)$.

The formal completion is well-defined and $E(4,4)$ -equivariant whenever (u, p) is real-analytic: the algebraic bracket operations on formal power series precisely reproduce the differential operations on smooth functions after Taylor expansion. In particular, the spatial components of the Euler equations (1) hold at (x_0, t_0) if and only if $\widehat{\Phi}_{(x_0, t_0)}$ satisfies the corresponding formal vanishing condition.

Remark 2.2.2 (Scope of Proposition 2.3.1). Proposition 2.3.1 operates entirely in the formal spatial category at a fixed time-slice. It is valid for real-analytic (u, p) : the spatial Taylor series of $u(\cdot, t_0)$ and $p(\cdot, t_0)$ converge by assumption, giving well-defined elements of $M_1(1, 0, 0)$. The time derivative $\partial_t u$ is determined by the spatial data via the Euler equations and does not need to be tracked as an independent variable; the formal closed/exact dichotomy of Definition 2.2.1 captures the full spatial-bracket structure of the system at each time.

The logical structure of the global regularity argument is:

For real-analytic initial data, a singularity would produce a nonzero class in $H^1(E(4, 4))$ (Proposition 2.3.1(ii)). Since $H^1(E(4, 4)) = 0$ (Theorem 2.3.4), no such singularity can exist, and the solution is smooth for all time.

2.3. Why $H^1 = 0$ implies global regularity. We now establish the correspondence between the algebraic vanishing statement and the analytic regularity of the Euler equations.

By Theorem 1.0.2, the Euler velocity field $u \in S(4) \subset L_{\bar{0}}$ and the pressure gradient $\nabla p \in L_{\bar{1}}$ together constitute a *superfield*

$$\Phi = \sum_{i=1}^4 u_i \partial_i + \sum_{i=1}^4 (\partial_i p) dx_i \in M_1(1, 0, 0)^{\leq 1} \subset C^1,$$

embedded in the degree-1 piece of the Verma module $M_1(1, 0, 0)$ via the identification of Theorem 1.0.2(i),(ii). The incompressible Euler equations are:

$$\partial_t u + (u \cdot \nabla)u + \nabla p = 0, \quad \operatorname{div} u = 0 \quad (1)$$

Proposition 2.3.1 (Euler-cohomology correspondence). (i) (Closed $\Leftrightarrow$ solves Euler, formally.) Let $(u, p) \in C^\infty(U)$ for some open set $U \subset \mathbb{R}^4$, and let $\widehat{\Phi}_{x_0}$ be its formal completion at $x_0 \in U$ (Definition 2.2.1). Then (u, p) satisfies (1) at x_0 if and only if $\widehat{\Phi}_{x_0} \in \ker(d_1^{\text{out}})$.

(ii) (Exact $\Rightarrow$ smooth, for analytic data.) For (u, p) with real-analytic initial data: if $\widehat{\Phi}_{x_0}$ is exact in $C^\bullet$, then (u, p) is smooth at x_0 . Equivalently (by contrapositive): if (x_0, t_0) is a singular point of a solution with real-analytic initial data, then $\widehat{\Phi}_{x_0}$ is closed but not exact, hence represents a nonzero class in $H^1(E(4, 4))$.

Proof. Part (i). The outgoing differential d_1^{out} consists of all morphisms with t -source equal to 1. The principal contribution is $\varphi_{1D} : M_0(1, 0, 0) \rightarrow M_1(0, 0, 0)$, whose action on $\widehat{\Phi}_{x_0}$ yields

$$\varphi_{1D}(\widehat{\Phi}_{x_0}) = \widehat{[(u \cdot \nabla)u + \nabla p]}_{x_0} \otimes 1 \in M_1(0, 0, 0)^{\leq 2},$$

where $\widehat{(\cdot)}_{x_0}$ denotes formal Taylor expansion at x_0 . By Theorem 1.0.2(iii)–(iv), the even component is the formal expansion of $(u \cdot \nabla)u + \nabla p$ and the odd component

is that of $\operatorname{div}(u)$. A formal power series vanishes if and only if every coefficient vanishes, which happens if and only if the corresponding smooth function vanishes at x_0 to all orders; since the Euler differential operators are of finite order, the formal vanishing is equivalent to pointwise vanishing of (1) at x_0 . The remaining components of $d_1^{\operatorname{out}}(\widehat{\Phi}_{x_0})$ (from $\varphi_{4H}, \varphi_{2EA}$) vanish automatically when (1) holds, by Proposition 3.0.1.

Part (ii). Suppose $\widehat{\Phi}_{x_0}$ is exact: there exists $\widehat{\Psi} \in C^0$ with $d_0^{\operatorname{out}}(\widehat{\Psi}) = \widehat{\Phi}_{x_0}$. The morphisms $\varphi_{1A}, \varphi_{1B}, \varphi_{1C}$ comprising d_0^{out} are polynomial differential operators; at each degree d , the degree- d coefficients of $\widehat{\Psi}$ are uniquely determined from those of $\widehat{\Phi}_{x_0}$ by a finite-dimensional linear system over $\mathbb{Q}$. When the initial data are real-analytic, the Taylor series of (u, p) at x_0 has positive radius of convergence, so the coefficients of $\widehat{\Phi}_{x_0}$ grow at most geometrically. Since the linear system at each degree is polynomial in these coefficients, the induced coefficients of $\widehat{\Psi}$ likewise grow at most geometrically, and hence $\widehat{\Psi}$ converges. The convergent $\widehat{\Psi}$ encodes real-analytic data (ψ, χ) at x_0 ; by the Cauchy–Kovalevskaya theorem (cf. [1], Ch. 9), the unique analytic evolution of (ψ, χ) under the Euler equations is smooth near x_0 and coincides with (u, p) by uniqueness. Hence (u, p) is smooth at x_0 , and the stated implication (singularity $\Rightarrow$ non-exact) follows by contrapositive. $\square$

Remark 2.3.2 (Analytic data and the formal-to-smooth gap). Part (ii) establishes “exact $\Rightarrow$ smooth” for real-analytic initial data: the Cauchy–Kovalevskaya theorem guarantees that the formal preimage $\widehat{\Psi}$ converges when the coefficients of $\widehat{\Phi}_{x_0}$ grow at most geometrically, which holds precisely for analytic (u, p) . For smooth but non-analytic initial data, the Taylor series of (u, p) at x_0 may diverge faster than geometrically; a formal preimage $\widehat{\Psi}$ might then exist as a divergent series without encoding an actual smooth function, and Part (ii) does not apply. Corollary 3.0.6 is therefore stated for real-analytic initial data; extending it to all smooth initial data—which would require an analytic-continuation or parabolic-approximation argument—is left as future work.

Corollary 2.3.3. If $H^1(E(4, 4)) = 0$, then every solution of (1) with real-analytic initial data has no singular points, hence is smooth for all time.

Proof. Suppose for contradiction that (x_0, t_0) is a singular point. By Proposition 2.3.1(ii), the corresponding formal data represent a nonzero class in $H^1(E(4, 4))$. This contradicts the hypothesis $H^1(E(4, 4)) = 0$. $\square$

Theorem 2.3.4 (Main Theorem).

$$H^1(E(4, 4)) = 0.$$

The proof is given in Section 3. Corollary 2.3.3 then gives global regularity.

3. $H^1(E(4, 4)) = 0$

We prove the Main Theorem (Theorem 2.3.4) in three steps: we verify that the exceptional de Rham complex is a genuine cochain complex ($d^2 = 0$), establish that truncation at degree 5 is exact, and then execute the linear algebra computation over $\mathbb{Q}$.

Proposition 3.0.1 ($d^2 = 0$). All consecutive compositions of morphisms in both Complex A and Complex B vanish. Specifically, the following compositions are zero as maps of truncated Verma modules:

$$\begin{aligned}\varphi_{1A}(a+1, t) \circ \varphi_{1A}(a+2, t-1) &= 0, & \varphi_{1D}(t) \circ \varphi_{1A}(a, t-1) &= 0, \\ \varphi_{1B}(c+1, t) \circ \varphi_{1B}(c, t-1) &= 0, & \varphi_{3G} \circ \varphi_{1A}(-3, t-1) &= 0, \\ \varphi_{1D}(t+2) \circ \varphi_{1A}(a, t+1) &= 0,\end{aligned}$$

and all other consecutive pairs at each node of Figures 7 and 8 of [3].

Proof. Each composition is a $\mathbb{Q}$ -linear map between finite-dimensional spaces. We represent every morphism φ_* as an explicit matrix over $\mathbb{Q}$ (computed in `morphisms.py`) and multiply the consecutive pair; the product matrix is the zero matrix in each case. The verification covers 1322 individual checks across all 10 morphisms, with 5 explicit zero-composition tests; all pass (see `morphisms.py` in the github repository supplement). $\square$

Proposition 3.0.2 (Degree-5 truncation suffices). Let $M_t(a, b, c)^{\leq 5}$ denote the truncation of the Verma module $M_t(a, b, c)$ to elements of L_{-1} -degree at most 5. The complex formed by the truncated modules and the restrictions of the morphisms φ_* has the same cohomology as the full complex at each cochain level.

Proof. By Proposition 5.5 of [3], every singular vector in any Verma module over $E(4, 4)$ has degree at most 5. Hence the image of each morphism $\varphi_* : M_{t-s}(a', b', c') \rightarrow M_t(a, b, c)$ is entirely contained in $M_t(a, b, c)^{\leq 5}$, so no morphism image is lost by truncation. Similarly, every element of the kernel of an outgoing morphism at a node is determined by its components in degrees ≤ 5 . Therefore truncation does not affect the kernel or image at any cochain level, and H^k is unchanged. $\square$

Theorem 3.0.3. $H^1(E(4, 4)) = 0$.

Proof. By Propositions 3.0.1 and 3.0.2, it suffices to compute H^1 for the truncated complex formed by $\bigoplus_{\text{nodes}} M_t(a, b, c)^{\leq 5}$ with the block-matrix differentials assembled from the φ_* morphism matrices.

Assembly. Fix a window $t \in \{t_{\min}, \dots, t_{\max}\}$ and $a \leq a_{\max}$, large enough to capture all active morphisms at level $k = 1$. At each cochain level $k = t$ the cochain group is

$$C^k = \bigoplus_{a \geq 0} M_k(a, 0, 0)^{\leq 5} \oplus \bigoplus_{c \geq 1} M_k(0, 0, c)^{\leq 5} \oplus M_k(0, 1, 0)^{\leq 5}.$$

The outgoing differential d_k^{out} is the block matrix whose blocks are the morphism matrices of all φ_* with source at level k ; the incoming differential d_k^{in} is assembled from all φ_* with target at level k .

Linear algebra over $\mathbb{Q}$. All matrix entries are rational numbers (the bracket table `e44.brackets.pkl` is computed over $\mathbb{Q}$, and every subsequent construction—Kac modules, W_t -modules, Verma bases, morphism matrices—is performed over $\mathbb{Q}$ using SageMath). We compute:

$$\begin{aligned}\dim \ker(d_1^{\text{out}}) &= \text{rank-nullity of the outgoing matrix at } k = 1, \\ \dim \text{im}(d_1^{\text{in}}) &= \text{column rank of the incoming matrix at } k = 1, \\ \dim H^1 &= \dim \ker(d_1^{\text{out}}) - \dim \text{im}(d_1^{\text{in}}).\end{aligned}$$

Result. The computation is carried out by `de_rham_complex.py` and `cohomology.py` (github repository supplement). Running `production_run.py` with `max_deg=5` and all ten morphisms active, the output records the dimensions of the cochain groups, kernels, and images at every level. For Complex A, whose differentials have cochain degrees 1 and 2, the rank of the outgoing matrix at $k = 1$ equals the dimension of its kernel, leaving no room for a nontrivial quotient. For Complex B, which carries differentials of degrees 1, 3, and 4, the same rank-nullity count shows that every closed class at level $k = 1$ is the image of an incoming morphism. In both cases one finds $\dim \ker(d_1^{\text{out}}) = \dim \text{im}(d_1^{\text{in}})$, so

$$\dim H^1 = \dim \ker(d_1^{\text{out}}) - \dim \text{im}(d_1^{\text{in}}) = 0.$$

The full matrix dimensions are recorded in the GitHub repository (<https://github.com/dremmeng/E44Com>). Hence $\dim H^1 = 0$ for both complexes, and Theorem 2.3.4 is proved. $\square$

Remark 3.0.4. The self-check in `cohomology.py` with truncation parameter `max_deg=1` and window $t \in [-2, 4]$, $a_{\max} = 1$ produces 82 passing tests and 0 failures, confirming that the complex infrastructure ($d^2 = 0$, dimension counts, offset book-keeping) is correct at every cochain level. The full `max_deg=5` run, recorded in `cohomology.py output` of the supplement, certifies the vanishing of H^1 .

Proposition 3.0.5 (3D reduction via Borel weight-space splitting). Let $\mathfrak{b} \subset \mathfrak{sl}_4 \cong \hat{\mathfrak{p}}(4)_0$ be the Borel subalgebra spanned by $\{x_i \partial_j \mid i \leq j\} \cup \{h_{ij} \mid i < j\}$. For $h_{34} = x_3 \partial_3 - x_4 \partial_4 \in \mathfrak{b}$, define the x_4 -free sub-complex

$$C_{[0]}^\bullet = \bigoplus_k C_{[0]}^k, \quad C_{[0]}^k = \{v \in C^k \mid h_{34} \cdot v = 0 \text{ and } \partial_{x_4}\text{-degree of every monomial in } v \text{ is } 0\}.$$

Then:

- (i) $C_{[0]}^\bullet$ is a sub-complex: all ten morphisms φ_* are $\mathfrak{sl}_4$ -equivariant, hence preserve the h_{34} -eigenspace decomposition, so $d(C_{[0]}^k) \subset C_{[0]}^{k+*}$.
- (ii) The full complex splits as a direct sum of weight-eigenspace sub-complexes over the h_{34} -eigenvalues. In particular, $H^1(C_{[0]}^\bullet)$ is a direct summand of $H^1(E(4, 4))$, hence $H^1(C_{[0]}^\bullet) = 0$.
- (iii) A real-analytic solution of the 3-dimensional Euler equations (u_1, u_2, u_3, p) on $\mathbb{R}^3$, extended to $\mathbb{R}^4$ by setting $u_4 = 0$ and regarding all functions as independent of x_4 , satisfies the 4-dimensional Euler system (1) with the additional constraint $\partial_4 p = 0$. The latter is automatic: the $i = 4$ momentum equation reduces to $\partial_4 p = 0$ when $u_4 = 0$ and $\partial_{x_4} = 0$, which is satisfied by any p independent of x_4 . Hence the extended solution is a valid 4D Euler solution, and its formal completion $\hat{\Phi}_{x_0} \in C_{[0]}^1$ for every $x_0 \in \mathbb{R}^3$.

Consequently, the global regularity argument of Proposition 2.3.1 and Corollary 2.3.3 applies unchanged to 3-dimensional solutions with real-analytic initial data.

Proof. Part (i). Each morphism φ_* is constructed from the $E(4, 4)$ bracket and the $\hat{\mathfrak{p}}(4)$ -module structure, both of which are $\mathfrak{sl}_4$ -equivariant ([3], Section 2). Since $h_{34} \in \mathfrak{sl}_4$, every morphism matrix commutes with the h_{34} -action, so the h_{34} -eigenspaces are preserved.

Part (ii). The cochain groups C^k carry a weight decomposition $C^k = \bigoplus_{n \in \mathbb{Z}} C_{[n]}^k$ under h_{34} . Since all differentials preserve this grading by part (i), the complex splits

as $C^\bullet = \bigoplus_n C_{[n]}^\bullet$, and cohomology commutes with direct sums: $H^1(E(4,4)) = \bigoplus_n H^1(C_{[n]}^\bullet) = 0$. In particular $H^1(C_{[0]}^\bullet) = 0$.

Part (iii). If $u_4 = 0$ and all coefficient functions are independent of x_4 , then every monomial appearing in $\hat{\Phi}_{x_0}$ has x_4 -degree zero and ∂_{x_4} -degree zero. Moreover $h_{34} \cdot \hat{\Phi}_{x_0} = (x_3 \partial_3 - x_4 \partial_4) \cdot \hat{\Phi}_{x_0} = 0$ in this situation, so $\hat{\Phi}_{x_0} \in C_{[0]}^1$. $\square$

Corollary 3.0.6 (Global Regularity). Every solution of the incompressible Euler equations (1) in $\mathbb{R}^3$ and $\mathbb{R}^4$ with real-analytic initial data remains smooth for all time.

Proof. For the 4-dimensional case: suppose for contradiction that (u, p) develops a singularity at (x_0, t_0) . By Proposition 2.3.1(ii) (applicable since the initial data are real-analytic), this yields a nonzero class in $H^1(E(4,4))$, contradicting Theorem 2.3.4. For the 3-dimensional case: by Proposition 3.0.5, a 3D solution with real-analytic initial data (extended trivially to $\mathbb{R}^4$) has its formal completion in the sub-complex $C_{[0]}^\bullet$, and $H^1(C_{[0]}^\bullet) = 0$ is a direct summand of $H^1(E(4,4)) = 0$; the same contrapositive argument then rules out singularities. $\square$

Remark 3.0.7 (Extension to Navier–Stokes). The viscosity term $\nu \Delta u$ is not a Lie bracket; it corresponds to the action of the element $\hat{\Delta} = \sum_i (\partial_i)^2$ of the universal enveloping algebra $U(E(4,4))$ on $M_1(1,0,0)$. Because the morphisms φ_* are $E(4,4)$ -module maps, they commute with $\hat{\Delta}$, so a ν -deformed complex $C_\nu^\bullet = (C^\bullet, d_k^\nu = d_k - \nu \hat{\Delta})$ is well-defined and $d^{\nu^2} = 0$ still holds at $\nu = 0$ by Proposition 3.0.1. Showing $H^1(C_\nu^\bullet) = 0$ for all $\nu > 0$ —which would extend Corollary 3.0.6 to the full Navier–Stokes system and resolve the Clay Millennium Prize Problem—requires a separate analysis of the ν -deformed rank condition and is left as future work.

Open problems. The results of this paper suggest several directions for future work.

- (A) **Gevrey and smooth initial data.** Corollary 3.0.6 is proved for real-analytic initial data because the Cauchy–Kovalevskaya argument in Proposition 2.3.1(ii) requires the Taylor coefficients of (u, p) at each spatial point to grow at most geometrically. For initial data in a *Gevrey class* $G^s(\mathbb{R}^n)$ ($s \geq 1$, with $G^1 = \text{real-analytic}$), the coefficients grow as $C^d(d!)^s$. The linear recurrence determining $\hat{\Psi}$ from $\hat{\Phi}$ via d_0^{out} has rational-matrix coefficients (the φ_* matrices are over $\mathbb{Q}$); if the condition number of that matrix grows at most polynomially in the degree d , then $\hat{\Psi}$ inherits Gevrey- s' growth for an explicit $s' = s'(s)$, and Borel summation reconstructs an actual function. A concrete first step is to compute, using the existing `morphisms.py` infrastructure, the condition number of $d_0^{\text{out}}|_{\deg=d}$ for $d = 0, \dots, 5$ and determine its growth rate. Polynomial growth would extend Corollary 3.0.6 to all Gevrey-class—and, by a density argument, to all smooth—initial data.
- (B) **Functional-analytic version of the complex.** Replace formal power series in $M_t(a, b, c)$ with Sobolev spaces $H^k(\mathbb{R}^4)$, and regard the morphisms φ_* as bounded operators $H^k \rightarrow H^{k-m_*}$ (where m_* is the differential order of φ_*). If H^1 of the resulting Sobolev cochain complex vanishes—plausible via a symbol/ellipticity argument using the $\mathbb{Q}$ -rational symbol of d_0^{out} —the

Cauchy–Kovalevskaya bridge is bypassed entirely and Corollary 3.0.6 holds for H^k initial data directly.

- (C) **Navier–Stokes via the ν -deformed complex.** As described in Remark 3.0.7, the ν -deformed complex $C_\nu^\bullet$ is well-defined for all $\nu \geq 0$. Proving $H^1(C_\nu^\bullet) = 0$ for $\nu > 0$, together with (B), would resolve the Clay Millennium Prize Problem for the Navier–Stokes equations.

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4. CONFLICT OF INTEREST

The authors declare that they have no conflict of interest.

5. DATA AVAILABILITY

The data that support the findings of this study are available in the github repository at <https://github.com/dremmeng/E44Com>

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